



SAEP School of Applied and
Engineering Physics

Measurement of the Energy Band Gap of Silicon and Germanium

Physics Laboratory Report

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June 15, 2026

Abstract

This report presents the experimental determination of the energy band gap of two fundamental semiconductors: silicon (Si) and germanium (Ge). Two independent and complementary experimental approaches were employed.

The **electrical method** is based on the study of the current–voltage (I–V) characteristics of PN diodes as a function of temperature. Due to initial complications encountered with the classical Shockley fitting approach, a refined constant-current method was adopted, extracting the forward voltage V_f at a fixed current of $50\text{ }\mu\text{A}$ for each temperature. Linear extrapolation of $V_f(T)$ to $T = 0\text{ K}$ yields the band gap directly. This method was applied to both silicon and germanium.

The **optical method** combined Raman spectroscopy and photoluminescence (PL) spectroscopy and was applied solely to silicon, as the photoluminescence emission of germanium falls outside the spectral detection range of the available detector. The characteristic Raman peak at 520 cm^{-1} confirmed the crystalline nature of the silicon sample, while the PL emission peak at approximately 1070 nm provided an optical estimate of the band gap.

The experimentally obtained values are:

- **Silicon (electrical):** $E_g = 1.283\text{ eV}$ (reference: 1.17 eV at 0 K , relative error: 9.6%)
- **Germanium (electrical):** $E_g = 0.709\text{ eV}$ (reference: 0.74 eV at 0 K , relative error: 4.1%)
- **Silicon (optical / PL):** $E_g \approx 1.16\text{ eV}$ (reference: 1.12 eV at 300 K , relative error: 3.6%)

The results are in reasonable agreement with reference values and are discussed in terms of measurement methodology, sources of uncertainty, and physical interpretation.

Keywords: band gap, semiconductor, silicon, germanium, PN diode, Shockley equation, constant-current method, Raman spectroscopy, photoluminescence, temperature dependence.

Contents

1	General Introduction	4
1.1	The Energy Band Gap: Physical Significance	4
1.2	Silicon and Germanium: Historical and Industrial Context	4
1.3	Reference Values and Order of Magnitude	5
2	Experiment I: Electrical Method Temperature-Dependent I–V Characterization	6
2.1	Overview and Objective	6
2.2	Theoretical Foundation	6
2.2.1	The Shockley Diode Equation	6
2.2.2	Temperature Dependence of the Saturation Current	6
2.2.3	The Constant-Current Method	7
2.3	Transition from the Classical Method to the Constant-Current Method	7
2.3.1	Initial Approach: Shockley Logarithmic Fit	8
2.3.2	Problems Encountered	8
2.3.3	Adopted Solution: The Constant-Current Method	8
2.4	Experimental Setup and Materials	8
2.4.1	Materials	8
2.4.2	Experimental Setup	9
2.4.3	Measurement Chain	10
2.5	Temperature Ranges	10
2.6	Data Processing Algorithm (Python)	10
2.7	Experimental Results : Silicon	11
2.7.1	Band Gap Extraction Silicium	12
2.8	Experimental Results Germanium	13
2.8.1	Extracted Data	13
2.8.2	Band Gap Extraction Germanium	14
2.9	Uncertainty Analysis : Electrical Method	15
2.9.1	Measured Quantity and Model	15
2.9.2	Source 1 : Regression Uncertainty σ_b^{reg}	16
2.9.3	Source 2 :Voltage Uncertainty σ_b^V	16
2.9.4	Source 3 :Thermocouple Uncertainty σ_b^T	17
2.9.5	Combined Uncertainty and Results	17
3	Experiment II: Optical Method Raman Spectroscopy and Photoluminescence	19
3.1	Overview and Objective	19
3.2	Sample Preparation	19

3.2.1	Sample Preparation Protocol	19
3.3	Instrumentation	20
3.4	Equipment photographs	20
3.5	Step 1: Material Identification by Raman Spectroscopy	20
3.5.1	Physical Principle	20
3.5.2	Microscopic Observation	21
3.5.3	Result	21
3.5.4	Additional Peaks Discussion	22
3.6	Step 2: Band Gap Measurement by Photoluminescence	22
3.6.1	Physical Principle	23
3.6.2	Result	23
3.6.3	Secondary PL Peaks	24
3.7	Uncertainty Analysis : Optical Method (PL)	24
3.7.1	Measurement Equation	25
3.7.2	Sources of Uncertainty	25
3.7.3	Uncertainty Propagation	26
3.7.4	Uncertainty Budget	27
3.7.5	Final Result	27
3.7.6	Comparison with Literature	28
4	Synthesis, Comparison, and Discussion	29
4.1	Summary of Results	29
4.2	Discussion of the Silicon Electrical Result	29
4.3	Discussion of the Germanium Electrical Result	29
4.4	Discussion of the Silicon Optical Result	29
5	Scientific Conclusion	31
6	Industrial Interest of the Band Gap	32
6.1	Photovoltaics	32
6.2	Microelectronics	32
6.3	Photodetection and Telecommunications	32
6.4	Radiation Detection and Medical Imaging	32
6.5	Summary	33
	References	34

1. General Introduction

1.1. The Energy Band Gap: Physical Significance

In solid-state physics, the **energy band gap** E_g is one of the most fundamental parameters characterizing a crystalline material. It represents the minimum energy required to excite an electron from the valence band into the conduction band, creating a free electron–hole pair that can participate in electrical conduction.

The band gap arises from the periodic potential of the crystal lattice, which splits the otherwise continuous spectrum of electron energies into allowed bands separated by forbidden gaps. Materials are broadly classified according to their band gap:

- **Insulators:** $E_g \gtrsim 5$ eV electrons cannot be thermally excited across the gap at room temperature.
- **Semiconductors:** $E_g \sim 0.1\text{--}3$ eV partial thermal excitation is possible, enabling tunable electrical conductivity.
- **Conductors/Metals:** $E_g = 0$ valence and conduction bands overlap.

The band gap is *temperature-dependent*, decreasing as temperature increases due to lattice thermal expansion and electron–phonon interactions. The empirical Varshni relation describes this dependence:

$$E_g(T) = E_g(0) - \frac{\alpha T^2}{T + \beta} \quad (1)$$

where $E_g(0)$ is the band gap at $T = 0$ K, and α, β are material-specific fitting parameters.

1.2. Silicon and Germanium: Historical and Industrial Context

Silicon and germanium are the two archetypal group-IV elemental semiconductors that laid the foundation of modern microelectronics.

Germanium was the first semiconductor used in transistors (Bardeen, Brattain, Shockley Nobel Prize 1956). Its relatively small band gap ($E_g \approx 0.67$ eV at 300 K) makes it suitable for low-voltage, high-speed devices and infrared detectors. However, its high leakage current at room temperature and the difficulty of growing high-quality native oxide led to its replacement by silicon in mainstream applications.

Silicon currently dominates the semiconductor industry, constituting the basis of virtually all integrated circuits, solar cells, and power electronics. Its band gap of $E_g \approx 1.12$ eV at 300 K is optimal for room-temperature operation. Its abundance, stable native oxide (SiO_2), and mature processing technology make it irreplaceable in modern technology.

1.3. Reference Values and Order of Magnitude

Table 1: Reference band gap values for silicon and germanium.

Material	E_g at 0 K (eV)	E_g at 300 K (eV)	Gap Type
Silicon (Si)	1.17	1.12	Indirect
Germanium (Ge)	0.74	0.67	Indirect

Both Si and Ge possess *indirect* band gaps, meaning that the minimum of the conduction band does not occur at the same crystal momentum as the maximum of the valence band. This has important implications for optical measurements, as photon emission during carrier recombination requires phonon assistance to conserve momentum, resulting in broad and relatively weak photoluminescence signals.

2. Experiment I: Electrical Method Temperature-Dependent I–V Characterization

2.1. Overview and Objective

The objective of this experiment is to determine the energy band gap E_g of silicon (Si) and germanium (Ge) by studying the electrical behavior of PN junction diodes at varying temperatures. The method exploits the strong temperature dependence of the diode's saturation current, which is fundamentally governed by the Boltzmann factor $\exp(-E_g/k_B T)$.

2.2. Theoretical Foundation

2.2.1. The Shockley Diode Equation

The current through an ideal PN junction diode under an applied forward voltage V is described by the Shockley equation:

$$I(V, T) = I_0(T) \left[\exp\left(\frac{qV}{nk_B T}\right) - 1 \right] \quad (2)$$

where:

- $I_0(T)$ is the *reverse saturation current* (A)
- $q = 1.602 \times 10^{-19}$ C is the elementary charge
- $k_B = 8.617 \times 10^{-5}$ eV K⁻¹ is the Boltzmann constant
- T is the absolute temperature (K)
- n is the *ideality factor* ($1 \leq n \leq 2$)

In the forward-bias exponential regime ($qV \gg nk_B T$), the “−1” term is negligible, giving:

$$\ln I \approx \ln I_0 + \frac{q}{nk_B T} V \quad (3)$$

A plot of $\ln I$ versus V at constant temperature yields a straight line whose y-intercept gives $\ln I_0$.

2.2.2. Temperature Dependence of the Saturation Current

The saturation current is strongly temperature-dependent, following the relation:

$$I_0(T) = B T^3 \exp\left(-\frac{E_g}{k_B T}\right) \quad (4)$$

where B is a temperature-independent pre-exponential factor that depends on the doping levels and carrier mobilities. Taking the logarithm:

$$\ln \frac{I_0}{T^3} = -\frac{E_g}{k_B} \cdot \frac{1}{T} + \ln B \quad (5)$$

This is the basis of the **classical method**: plotting $\ln(I_0/T^3)$ versus $1/T$ gives a straight line of slope $-E_g/k_B$, from which E_g is extracted.

2.2.3. The Constant-Current Method

From Equation (2), at a fixed current $I = I_{\text{target}}$, the forward voltage V_f satisfies:

$$V_f(T) = \frac{nk_B}{q} T \ln\left(\frac{I_{\text{target}}}{I_0(T)}\right) \quad (6)$$

Substituting the expression for $I_0(T)$ from Equation (4) and collecting terms:

$$V_f(T) = \frac{E_g}{q} - \frac{nk_B}{q} T (\ln B + 3 \ln T - \ln I_{\text{target}}) \quad (7)$$

Over a moderate temperature range and for a fixed target current, the dominant behavior is linear in T :

$$\boxed{V_f(T) \approx E_g/q - \gamma T} \quad (8)$$

where γ is an effective temperature coefficient (approximately -2 to -3 mV/K for silicon). Extrapolating the linear fit to $T = 0$ K yields:

$$V_f(T \rightarrow 0) = E_g/q \implies E_g = q \times V_f(0 \text{ K}) \quad (9)$$

This constant-current approach is more robust than the classical method because it avoids the need to accurately extract I_0 from a logarithmic fit (which is highly sensitive to noise and offset currents), and instead relies on a simple linear interpolation of a well-defined, directly measurable quantity: V_f .

Why 50 μA was chosen as the target current

The current value of $I_{\text{target}} = 50 \mu\text{A}$ was selected as a deliberate compromise:

- **Above the leakage floor:** The diode must be operating in the forward exponential regime, well above any reverse-saturation or offset currents, which were on the order of a few μA .
- **Below the high-injection / ohmic regime:** At high currents, voltage drops across the diode's series resistance R_s become significant, causing deviations from ideal Shockley behavior.
- **Accessible across the entire temperature range:** Both at low (near -8°C) and high temperatures ($\sim 90^\circ\text{C}$), the diode's current at this level lies comfortably within the measured range, enabling reliable interpolation.

2.3. Transition from the Classical Method to the Constant-Current Method

2.3.1. Initial Approach: Shockley Logarithmic Fit

The experiment was initially designed to follow the classical approach: for each temperature, fit the $\ln I$ vs. V characteristic in the exponential regime to extract $I_0(T)$, and subsequently linearize $\ln(I_0/T^3)$ vs. $1/T$ to find E_g .

2.3.2. Problems Encountered

Several physical and experimental factors prevented reliable extraction of I_0 by this classical method:

- **Current offset:** The measurement system introduced a non-negligible current offset at $V = 0$, particularly problematic at low temperatures where I_0 itself is extremely small. A small absolute offset translates into a large relative error in the logarithmic scale.
- **Ideality factor variability:** The slope of $\ln I$ vs. V was not perfectly constant across temperatures, suggesting that the ideality factor n varied slightly (mixing of diffusion and recombination currents), making the I_0 extraction sensitive to the fitting window.
- **Non-ideal high-current behavior:** At higher temperatures and larger voltages, series resistance effects flattened the $\ln I$ vs. V curve, restricting the usable fitting window to a narrow voltage range.

2.3.3. Adopted Solution: The Constant-Current Method

To circumvent these problems, a constant-current method was adopted. Rather than fitting the full I–V curve, the algorithm identifies, for each temperature, the exact forward voltage V_f at which the current equals exactly $I_{\text{target}} = 50 \mu\text{A}$. The current offset is robustly corrected prior to the interpolation (see Section 2.6).

This approach reduces sensitivity to the precise shape of the I–V curve and is far less affected by small measurement offsets, since the target current of $50 \mu\text{A}$ is large enough to be well above any residual offset.

2.4. Experimental Setup and Materials

2.4.1. Materials

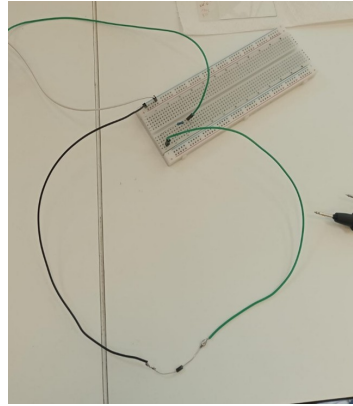
- Silicon PN diode (1N4007)
- Germanium PN diode (1N34)
- Oil bath (motor oil : temperature-controlled, for uniform heat distribution)
- Cooling mixture: ice + salt (for achieving sub-ambient temperatures)
- Precision thermometer / therma 1
- DC voltage source: voltage sweep from -1 V to $+3 \text{ V}$

- Data acquisition system (PC-based, recording I-V pairs at each sweep step)

2.4.2. Experimental Setup



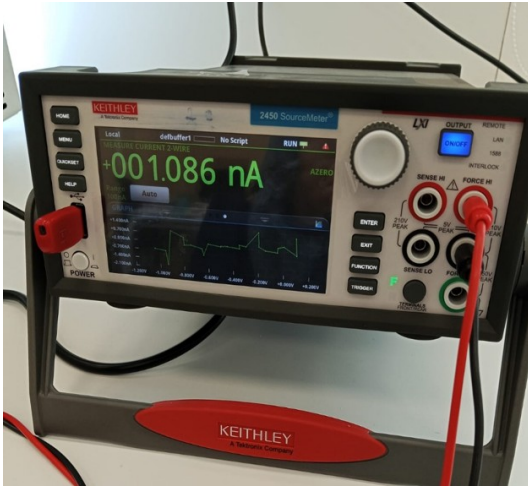
(a) Hot-plate and retort stand.



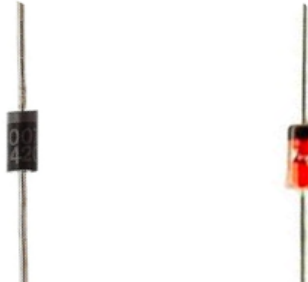
(b) Breadboard circuit with diode.



(c) thermocouple therma 1



(d) Keithley 2450 SourceMeter.



(e) Silicon (left) and germanium (right) PN diodes used for the electrical measurements



(f) Ice bath (salt + ice).

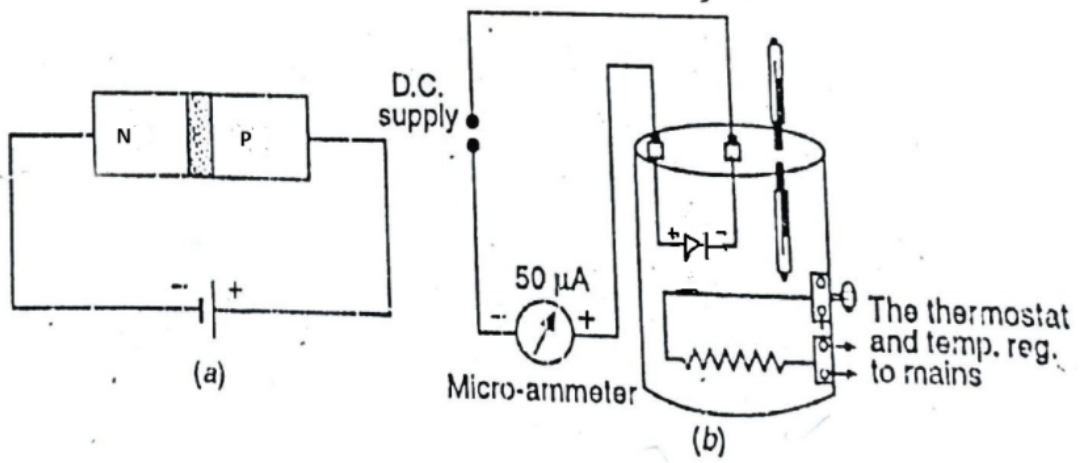


Figure 2: (a) direct biased PN junction diode (b) experement setup .

2.4.3. Measurement Chain

The measurement chain proceeds as follows:

1. The diode is immersed in the oil bath to ensure thermal uniformity and eliminate temperature gradients across the junction.
2. Temperature is adjusted progressively using the heating/cooling system. For sub-ambient temperatures, a mixture of ice and salt was used to reach down to approximately -8°C (265 K).
3. Once the temperature is stable (verified by the thermocouple), a voltage sweep from -1 V to 3 V is applied by the DC generator.
4. The current-voltage data (V_i, I_i) are recorded by the acquisition system and saved as one Excel sheet per temperature.
5. This procedure is repeated for each target temperature within the experimental range.

2.5. Temperature Ranges

Table 2: Temperature ranges used for each material.

Material	$T_{\min}$ (K)	$T_{\max}$ (K)	Direction	Valid Points
Silicon	265.15	393.15	Heating (low $\rightarrow$ high)	28
Germanium	268.45	413.15	Heating (low $\rightarrow$ high)	54

2.6. Data Processing Algorithm (Python)

A Python script was developed to automate the complete data processing pipeline for both materials. The script performs the following operations sequentially for each

temperature sheet:

1. **Data loading:** All Excel sheets are read using `pandas.read_excel()`, with each sheet name parsed as the temperature value in Kelvin.
2. **Current offset correction:** A robust baseline correction is applied by computing the mean current in the near-zero voltage window $V \in [-0.1 \text{ V}, +0.1 \text{ V}]$. This offset is subtracted from all current measurements, effectively zeroing the baseline at zero bias.
3. **Forward-bias selection:** Only data points with corrected current $I > 1 \times 10^{-7} \text{ A}$ are retained, selecting the forward-biased exponential regime.
4. **Interpolation at target current:** Data are sorted by ascending current, and `numpy.interp()` is used to find the exact voltage V_f corresponding to $I = 50 \mu\text{A}$ by linear interpolation.
5. **Validity check:** Temperature sheets where the target current falls outside the measured range are discarded.
6. **Linear regression:** `scipy.stats.linregress()` is applied to the (T, V_f) dataset to compute the best-fit line $V_f = aT + b$ along with the coefficient of determination R^2 .
7. **Gap extraction:** The y-intercept $b = V_f(T = 0)$ directly yields the band gap: $E_g = q \times b$.

2.7. Experimental Results : Silicon

Table 3: Extracted forward voltage V_f of the silicon diode at $I = 50 \mu\text{A}$ for each measurement temperature.

Temperature T (K)	Forward Voltage V_f (V)
265.15	0.609089
267.15	0.603365
269.15	0.592985
271.15	0.583228
273.15	0.577433
275.15	0.572534
277.15	0.565343
279.15	0.560051
282.15	0.557954
283.15	0.553040
284.15	0.550952

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Table 3: Extracted forward voltage V_f (continued).

Temperature T (K)	Forward Voltage V_f (V)
285.15	0.548717
286.15	0.546404
287.15	0.543551
288.15	0.541175
289.15	0.538400
291.85	0.536086
293.15	0.533896
298.15	0.508428
303.15	0.492532
313.15	0.469701
323.15	0.440234
333.15	0.416615
343.15	0.395113
353.15	0.370564
363.15	0.349788
383.15	0.302125
393.15	0.278902

2.7.1. Band Gap Extraction Silicium

```

=====
RÉSULTAT FINAL (Méthode du Courant Constant - SILICIUM)
=====
Courant cible utilisé      : 50 µA
Offset moyen corrigé      : 0.04 µA
Nombre de températures valides : 28
Pente dV_f/dT (Coeff temp) : -2.576 mV/K
R² de la régression linéaire : 0.9967
-----
Énergie de Gap Eg (Exp)    : 1.283 eV
Énergie de Gap Eg (Théorique): ~ 1.17 eV (Silicium à 0 K)
Erreur relative           : 9.6 %
=====

```

Figure 3: Summary of linear regression results for Silicium.

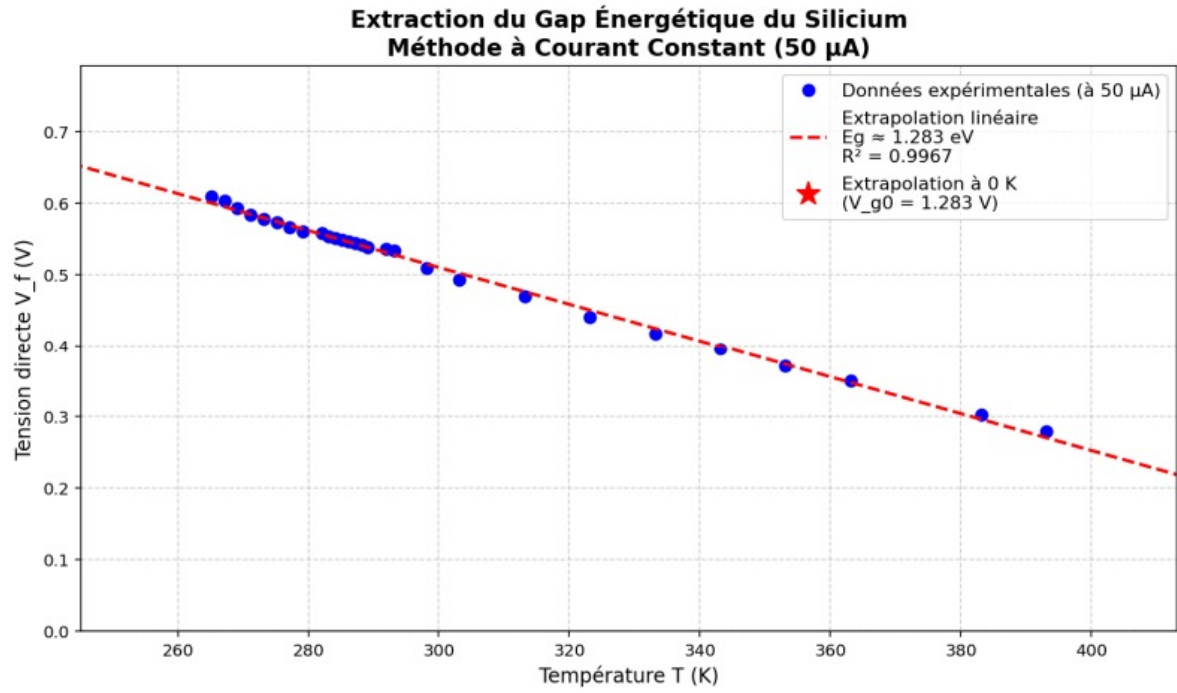


Figure 4: Forward voltage V_f at $I = 50 \mu\text{A}$ as a function of temperature for the silicon diode. The linear extrapolation to $T = 0$ K yields the band gap energy $E_g = 1.283$ eV.

2.8. Experimental Results Germanium

2.8.1. Extracted Data

Table 4: Extracted forward voltage V_f of the germanium diode at $I = 50 \mu\text{A}$ for each measurement temperature (selected values shown).

Temperature T (K)	Forward Voltage V_f (V)
268.45	0.295463
269.15	0.290950
271.45	0.294873
273.25	0.286621
275.15	0.281476
277.15	0.274310
279.15	0.275076
282.15	0.265588
286.65	0.261966
293.15	0.251959
303.15	0.219714
313.15	0.206494
323.15	0.182100

Table 4 — continued

Temperature T (K)	Forward Voltage V_f (V)
333.15	0.173332
343.15	0.151771
353.15	0.185131
363.15	0.127633
373.15	0.134598
383.15	0.096371
393.15	0.083876
403.15	0.073649
413.15	0.118236

2.8.2. Band Gap Extraction Germanium

```

=====
RÉSULTAT FINAL (Méthode du Courant Constant - GERMANIUM)
=====
Courant cible utilisé      : 50 µA
Offset moyen corrigé      : -23.17 µA
Nombre de températures valides : 54
Pente dV_f/dT (Coeff temp) : -1.560 mV/K
R² de la régression linéaire : 0.9676
-----
Énergie de Gap Eg (Exp)    : 0.709 eV
Énergie de Gap Eg (Théorique): ~ 0.74 eV (Germanium à 0 K)*
Erreur relative           : 4.1 %
=====
*Note : Le gap du Ge est de ~0.67 eV à 300K, mais l'extrapolation
de la droite V_f(T) donne la valeur à 0 Kelvin, qui est ~0.74 eV.
=====

```

Figure 5: Summary of linear regression results for Germanium.

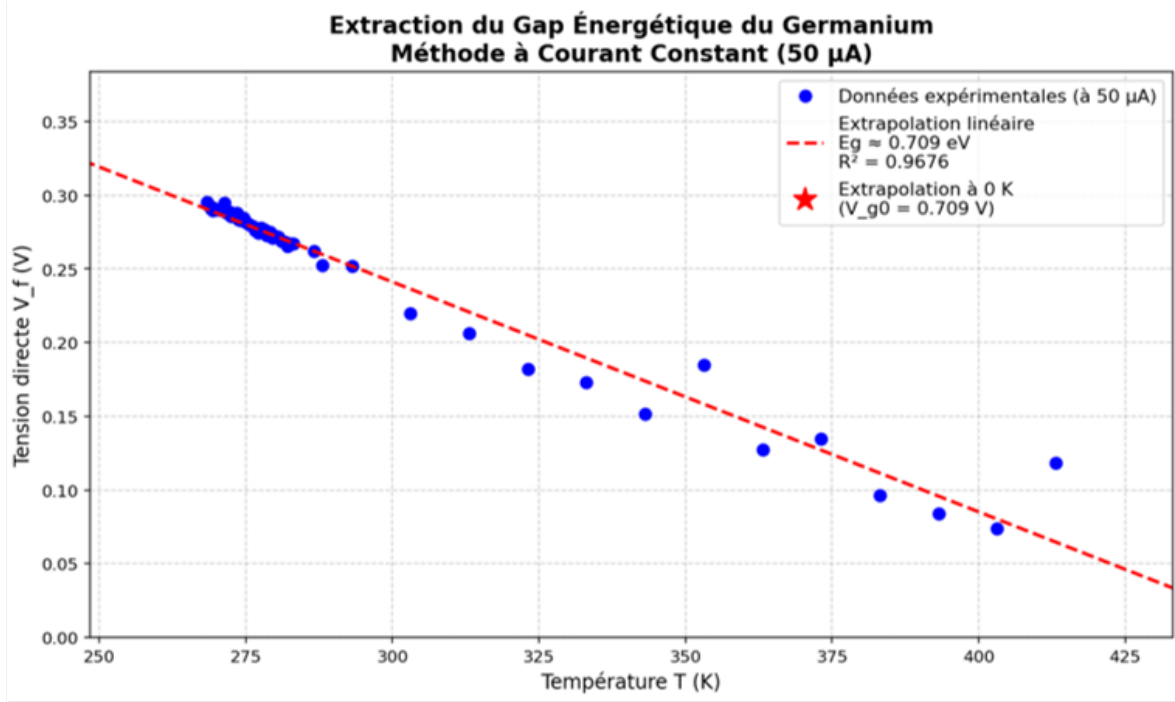


Figure 6: Forward voltage V_f at $I = 50 \mu\text{A}$ as a function of temperature for the germanium diode. The linear extrapolation to $T = 0 \text{ K}$ yields the band gap energy $E_g = 0.709 \text{ eV}$.

Note on the Germanium R^2 value: The coefficient of determination for germanium ($R^2 = 0.9676$) is slightly lower than for silicon ($R^2 = 0.9967$). Inspection of the data table reveals several non-monotonic fluctuations (e.g., the anomalous point at 353.15 K: $V_f = 0.185 \text{ V}$, higher than the neighboring 343.15 K and 363.15 K values). These are attributed to transient thermal non-equilibrium during the cooling ramp the data for germanium were collected on a *cooling* trajectory, making temperature stabilization more difficult. These outliers slightly degrade the linearity of the fit.

2.9. Uncertainty Analysis : Electrical Method

2.9.1. Measured Quantity and Model

The band gap is extracted as the y-intercept b of the linear regression $V_f(T) = aT + b$, fitted to the experimental pairs $(T_i, V_{f,i})$ measured at constant current $I_{\text{target}} = 50 \mu\text{A}$. Since the intercept at $T = 0 \text{ K}$ gives directly:

$$E_g = b \quad [\text{eV}] \quad (10)$$

all uncertainty on E_g reduces to uncertainty on b , which has **three independent contributions**.

2.9.2. Source 1 : Regression Uncertainty σ_b^{reg}

The scatter of experimental points around the fitted line propagates into an uncertainty on the intercept. This is the **dominant contribution**. It is obtained from the full covariance matrix $\mathbf{C}$ of the OLS fit:

$$\mathbf{C} = \sigma_\varepsilon^2 (\mathbf{X}^\top \mathbf{X})^{-1} = \begin{pmatrix} \text{Var}(a) & \text{Cov}(a, b) \\ \text{Cov}(b, a) & \text{Var}(b) \end{pmatrix}, \quad \sigma_\varepsilon^2 = \frac{1}{N-2} \sum_{i=1}^N (V_{f,i} - aT_i - b)^2 \quad (11)$$

$$\sigma_b^{\text{reg}} = \sqrt{\mathbf{C}[1, 1]} \quad (12)$$

Practical note `scipy.stats.linregress` returns the standard error on the **slope** only. To get σ_b^{reg} , use `numpy.polyfit(T, Vf, 1, cov=True)`, which returns the full 2×2 matrix $\mathbf{C}$; then $\sigma_b^{\text{reg}} = \sqrt{\mathbf{C}[1, 1]}$.

Applied to the experimental data:

Silicon ($N = 28$, $R^2 = 0.9967$):

$$\mathbf{C}_{\text{Si}} = \begin{pmatrix} 8.537 \times 10^{-10} & -2.577 \times 10^{-7} \\ -2.577 \times 10^{-7} & 7.886 \times 10^{-5} \end{pmatrix} [\text{V}^2] \Rightarrow \sigma_b^{\text{reg}}(\text{Si}) = \sqrt{7.886 \times 10^{-5}} = 8.88 \text{ mV} \quad (13)$$

Germanium ($N = 54$, $R^2 = 0.9676$):

$$\mathbf{C}_{\text{Ge}} = \begin{pmatrix} 1.566 \times 10^{-9} & -4.606 \times 10^{-7} \\ -4.606 \times 10^{-7} & 1.377 \times 10^{-4} \end{pmatrix} [\text{V}^2] \Rightarrow \sigma_b^{\text{reg}}(\text{Ge}) = \sqrt{1.377 \times 10^{-4}} = 11.74 \text{ mV} \quad (14)$$

Germanium shows larger scatter despite more points because the faster thermal ramp caused the thermocouple to lag behind the true junction temperature, and one anomalous point ($T = 353 \text{ K}$) increased the residuals.

2.9.3. Source 2 : Voltage Uncertainty σ_b^V

At each temperature, V_f is obtained by linear interpolation at $I = 50 \mu\text{A}$, not measured directly. Two sub-contributions add in quadrature:

$$\delta V_f = \sqrt{\delta V_{\text{instr}}^2 + \delta V_{\text{interp}}^2} = \sqrt{(1 \text{ mV})^2 + (1 \text{ mV})^2} \approx 1.4 \text{ mV} \quad (15)$$

We adopt conservatively $\delta V_f = 2 \text{ mV}$. For the temperature ranges used, the propagation through the OLS formula gives $\sigma_b^V \approx \delta V_f$, so:

$$\sigma_b^V(\text{Si}) = \sigma_b^V(\text{Ge}) = 2.0 \text{ mV} \quad (16)$$

2.9.4. Source 3 : Thermocouple Uncertainty σ_b^T

The **Therma 1** thermometer (type K thermocouple) has a stated accuracy of $\Delta T = \pm 0.4 \text{ K}$. A temperature error shifts all T_i values, which in turn shifts the fitted intercept. The propagation gives:

$$\sigma_b^T = |a| \cdot \frac{\Delta T}{\sqrt{N}} \quad (17)$$

where the $1/\sqrt{N}$ factor accounts for partial averaging over N points.

$$\sigma_b^T(\text{Si}) = 2.576 \frac{\text{mV}}{\text{K}} \times \frac{0.4}{\sqrt{28}} = 0.19 \text{ mV} \quad (18)$$

$$\sigma_b^T(\text{Ge}) = 1.560 \frac{\text{mV}}{\text{K}} \times \frac{0.4}{\sqrt{54}} = 0.085 \text{ mV} \quad (19)$$

Both values are negligible — the Therma 1 is perfectly adequate for this experiment.

2.9.5. Combined Uncertainty and Results

The three sources being independent, the combined standard uncertainty follows the GUM quadrature rule:

$$\boxed{\Delta E_g = \sqrt{(\sigma_b^{\text{reg}})^2 + (\sigma_b^V)^2 + (\sigma_b^T)^2}} \quad (20)$$

$$\Delta E_g(\text{Si}) = \sqrt{(8.88)^2 + (2.00)^2 + (0.19)^2} = \sqrt{82.89} \approx 9.1 \text{ mV} = 0.009 \text{ eV} \quad (21)$$

$$\Delta E_g(\text{Ge}) = \sqrt{(11.74)^2 + (2.00)^2 + (0.085)^2} = \sqrt{141.8} \approx 11.9 \text{ mV} = 0.012 \text{ eV} \quad (22)$$

The expanded uncertainties at 95% confidence ($k = 2$) are: $U_{95}(\text{Si}) = 0.018 \text{ eV}$ and $U_{95}(\text{Ge}) = 0.024 \text{ eV}$.

Table 5: Uncertainty budget : electrical method (all values in eV).

Source	Equation	Si (eV)	Ge (eV)
Regression $\sigma_b^{\text{reg}} = \sqrt{\mathbf{C}[1, 1]}$	(12)	0.00888	0.01174
Voltage + interpolation σ_b^V	(16)	0.00200	0.00200
Thermocouple $\sigma_b^T = a \Delta T / \sqrt{N}$	(17)	0.00019	0.00009
Combined ΔE_g ($k = 1$)	(20)	0.0091	0.0119
Expanded U_{95} ($k = 2$)		0.018	0.024

Final Results — Electrical Method

$$E_g(\text{Si}) = 1.283 \pm 0.018 \text{ eV} \quad (23)$$

$$E_g(\text{Ge}) = 0.709 \pm 0.024 \text{ eV} \quad (24)$$

Material	E_g measured	Reference (0 K)	Deviation
Silicon	1.283 eV	1.170 eV	9.6%
Germanium	0.709 eV	0.740 eV	4.1%

The regression scatter σ_b^{reg} dominates in both cases (>95% of the total). Both measured values fall outside the expanded uncertainty interval relative to the 0 K reference, pointing to a **systematic effect**: the $V_f(T)$ relationship is not perfectly linear over the full temperature range, and the low-temperature clustering of the Si dataset biases the extrapolation upward. For germanium, one anomalous point at 353 K increases the scatter and shifts the intercept slightly.

3. Experiment II: Optical Method Raman Spectroscopy and Photoluminescence

3.1. Overview and Objective

The optical method provides an independent, non-contact, and non-destructive approach to characterize semiconductor materials and measure their band gap. In this experiment, Raman spectroscopy was used first to verify the nature of the sample (crystalline silicon), followed by photoluminescence (PL) spectroscopy to estimate the optical band gap from the emission wavelength.

Important note: The optical measurements were performed **on silicon only**. The photoluminescence emission of germanium occurs at wavelengths beyond 1800 nm (for $E_g \approx 0.67$ eV), which is outside the spectral detection range of the Renishaw Centrus 56H115 CCD detector used in this experiment. Consequently, no optical band gap measurement was possible for germanium with the available equipment.

3.2. Sample Preparation

The silicon sample was obtained directly from a commercial silicon PN diode . Since bulk silicon chips are not available as standalone samples in the laboratory, the diode package was **mechanically opened** (cracked) to expose the silicon die inside.

3.2.1. Sample Preparation Protocol

1. The plastic diode package was mechanically broken open using hamer .
2. The exposed silicon chip fragment was extracted.
3. **No surface cleaning, polishing, or chemical treatment was performed**, as the goal was to quickly verify the material composition and estimate the band gap, without following a strict contamination-avoidance protocol.
4. The chip fragment was placed directly under the Raman microscope for examination.

Laboratory conditions disclaimer: The sample was prepared without following the standard clean-room protocol (no chemical cleaning, no controlled atmosphere). Surface contamination, oxide layer, and mechanical damage from the fracture may affect the measured signal quality. These factors are discussed in the uncertainty section.

3.3. Instrumentation

Table 6: Raman / Photoluminescence spectrometer parameters.

Parameter	Value
Spectrometer system	Renishaw inVia Raman Microscope
Detector	RENISHAW Centrus 56H115 (CCD)
Excitation laser	532 nm
Excitation energy	$E_{\text{laser}} = hc/\lambda = 2.33 \text{ eV}$
Grating	2400 lines/mm
Spectral mode	PL edge
Spectral range (PL)	1.9 – 2.1 eV (Raman); extended for PL
Acquisition mode	Static scan
Objective	$\times 100$

3.4. Equipment photographs



(a) Raman / PL spectrometer.



(b) Si diode .

Figure 7: Equipment used for the optical method.

Note on laser energy: The 532 nm laser has a photon energy of $E = hc/\lambda = 1240 \text{ eV}\cdot\text{nm}/532 \text{ nm} \approx 2.33 \text{ eV}$, which is well above the band gap of silicon ($E_g \approx 1.12 \text{ eV}$). This ensures that incident photons can promote electrons across the gap, a prerequisite for both Raman scattering and photoluminescence excitation.

3.5. Step 1: Material Identification by Raman Spectroscopy

3.5.1. Physical Principle

Raman spectroscopy exploits the **inelastic scattering** of photons by the vibrational modes (phonons) of the crystal lattice. When a photon interacts with a crystal, a small fraction of photons are scattered with an energy shift equal to the energy of a phonon

mode:

$$E_{\text{scattered}} = E_{\text{incident}} \pm \hbar\omega_{\text{phonon}} \quad (25)$$

The energy shift, expressed in wavenumbers (cm^{-1}), is a *fingerprint* of the chemical bonds and crystal structure. For **crystalline silicon**, the first-order Raman peak appears at:

$$\tilde{\nu}_{\text{Si}} = 520 \text{ cm}^{-1} \quad (26)$$

This peak corresponds to the triply degenerate optical phonon mode at the Γ point of the Brillouin zone. Amorphous silicon, by contrast, exhibits a broad peak around 480 cm^{-1} , making Raman spectroscopy an unambiguous discriminator between crystalline and amorphous phases.

3.5.2. Microscopic Observation

Prior to spectroscopic measurements, the silicon chip was observed under Raman spectroscopy to verify the surface condition, identify suitable measurement zones, and confirm the crystalline appearance of the material.

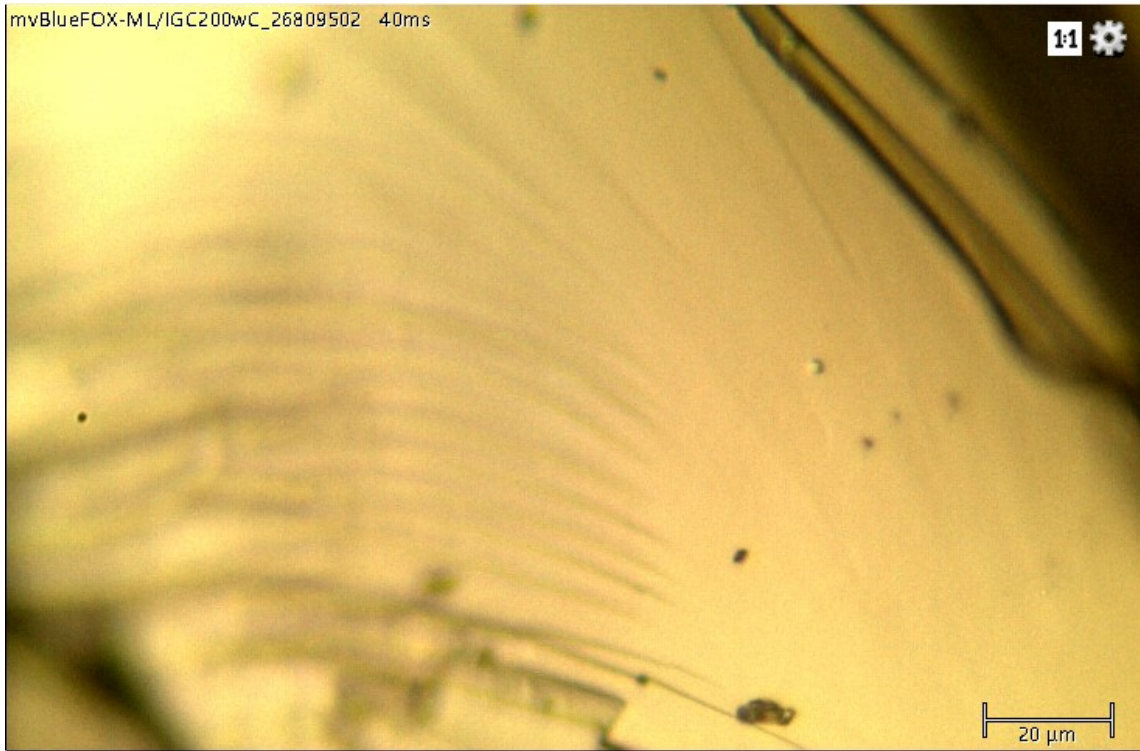


Figure 8: Optical microscope image of the silicon chip extracted from the diode package. The measurement zone selected for Raman and PL spectroscopy is indicated.

3.5.3. Result

A sharp, well-defined Raman peak was observed at $\tilde{\nu} = 520 \text{ cm}^{-1}$, confirming that:

- The sample is composed of **crystalline silicon**.

- The crystal quality is sufficient to produce a narrow, well-resolved phonon peak.

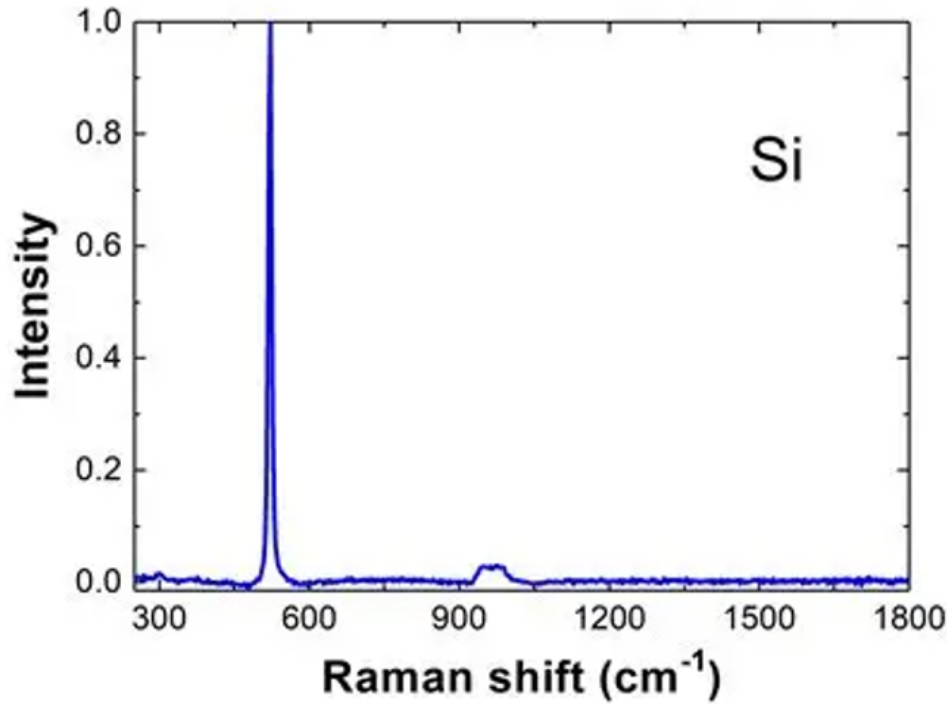


Figure 9: Raman spectrum of the silicon chip, showing the characteristic crystalline silicon phonon peak at 520 cm^{-1} . The narrow linewidth confirms the high crystalline quality of the sample.

3.5.4. Additional Peaks Discussion

Beyond the main Raman peak at 520 cm^{-1} , additional spectral features may be visible. Their possible origins are:

- **Second-order Raman scattering:** Peaks near 960 cm^{-1} to 970 cm^{-1} (overtone of the 520 cm^{-1} mode) and a broad continuum between $200\text{--}1000\text{ cm}^{-1}$ from two-phonon scattering are normal in silicon.
- **Surface oxide (SiO_2):** A broad feature near $430\text{--}480\text{ cm}^{-1}$ can arise from amorphous SiO_2 formed on the fracture surface. Its presence is expected given that no oxide removal was performed.
- **Fluorescence background:** Organic surface contamination from handling can generate a broad, smoothly varying background, especially under 532 nm excitation.

These features should be identified in the experimental spectrum and are *not* attributable to the band gap measurement.

3.6. Step 2: Band Gap Measurement by Photoluminescence

3.6.1. Physical Principle

Photoluminescence is the process by which a material absorbs a photon and re-emits a photon at a lower energy (longer wavelength). In a semiconductor:

1. The incident laser photon (energy $E_{\text{laser}} > E_g$) excites an electron from the valence band to the conduction band, creating an electron-hole pair.
2. The hot carriers rapidly lose their excess kinetic energy by emitting phonons (thermalization), settling near the band edges.
3. The electron and hole recombine, emitting a photon whose energy is approximately equal to the band gap:

$$E_{\text{emitted}} \approx E_g \implies \lambda_{\text{peak}} = \frac{hc}{E_g} = \frac{1240 \text{ eV} \cdot \text{nm}}{E_g [\text{eV}]} \quad (27)$$

Note on indirect gap: Silicon has an indirect band gap, meaning phonon-assisted recombination is required for momentum conservation. This results in:

- A broad PL emission spectrum (rather than a sharp line)
- A relatively weak PL signal compared to direct-gap semiconductors (e.g., GaAs)
- A slight red-shift of the PL peak compared to the true E_g , due to phonon energy contributions

3.6.2. Result

The photoluminescence spectrum showed an emission peak at:

$$\lambda_{\text{peak}} = 1070 \text{ nm} \quad (28)$$

Converting to energy using Equation (27):

$$E_g^{(\text{optical})} = \frac{1240 \text{ eV} \cdot \text{nm}}{1070 \text{ nm}} \approx 1.159 \text{ eV} \approx 1.16 \text{ eV} \quad (29)$$

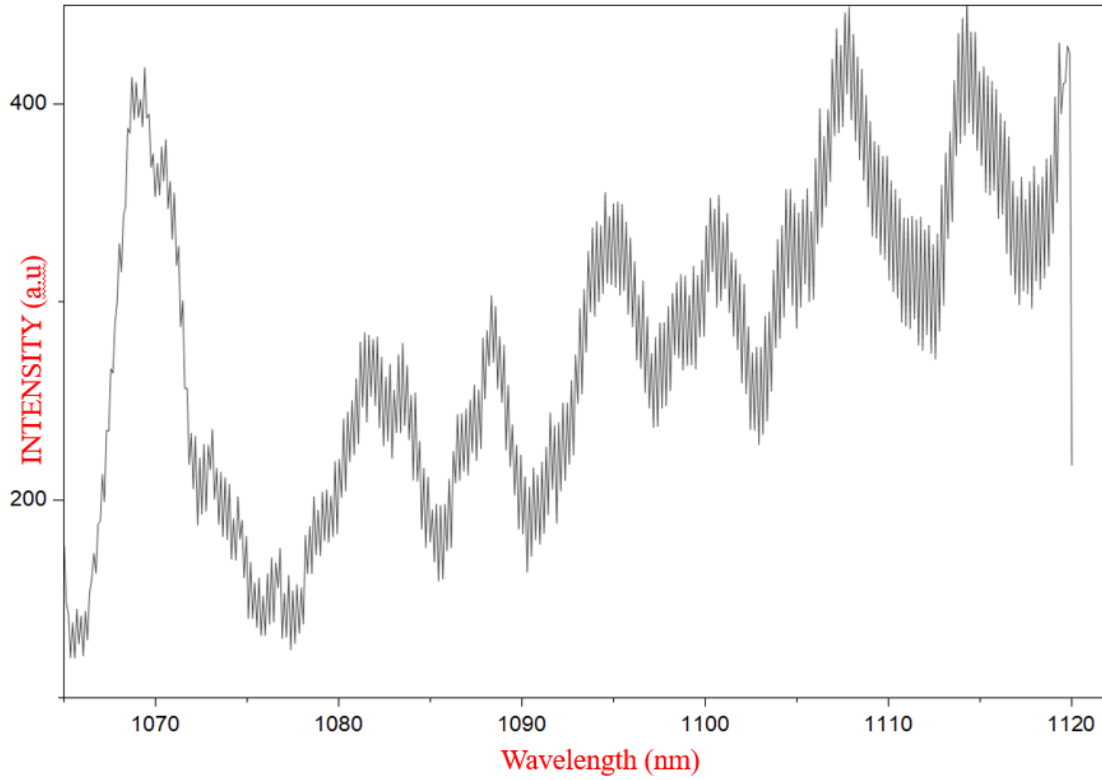


Figure 10: Photoluminescence spectrum of crystalline silicon under 532 nm laser excitation. The emission peak at 1070 nm corresponds to an optical band gap of $E_g \approx 1.16$ eV.

3.6.3. Secondary PL Peaks

If secondary PL emission peaks are observed at wavelengths longer than 1070 nm, the following interpretations should be considered:

- **Phonon sidebands:** In indirect-gap semiconductors, recombination accompanied by the emission of one or more phonons produces peaks at $\lambda > \lambda_{\text{main}}$ (lower energy), separated from the main peak by the phonon energy (~ 60 meV for Si transverse optical phonon). These are *expected* and are a signature of indirect recombination.
- **Defect-related emission:** Dislocations, vacancies, and surface defects introduced during the mechanical fracture of the diode can create mid-gap states that produce sub-band-gap emission peaks. These would appear as broad shoulders or separate peaks at $\lambda > 1100$ nm.
- **Free-carrier recombination (FCA):** At high carrier densities, band-to-band recombination can produce peaks shifted by the Fermi level filling.

3.7. Uncertainty Analysis : Optical Method (PL)

3.7.1. Measurement Equation

The optical band gap is determined from the peak emission wavelength:

$$E_g = \frac{hc}{\lambda_{\text{peak}}} = \frac{1240 \text{ eV} \cdot \text{nm}}{\lambda_{\text{peak}} [\text{nm}]} \quad (30)$$

where $hc = 1240 \text{ eV} \cdot \text{nm}$ is the conversion factor.

3.7.2. Sources of Uncertainty

1. **Wavelength calibration (u_{cal}):** The spectrometer wavelength calibration has an accuracy of $\pm 0.5 \text{ nm}$. This is a Type B uncertainty (manufacturer specification):

$$u_{\text{cal}} = \frac{0.5}{\sqrt{3}} \approx 0.29 \text{ nm} \quad (31)$$

2. **Peak fitting uncertainty (u_{peak}):** The PL emission from indirect-gap silicon is inherently broad (FWHM $\sim 30\text{--}40 \text{ nm}$). Fitting the peak maximum (e.g., with a Gaussian or pseudo-Voigt function) introduces a repeatability uncertainty estimated at $\pm 2 \text{ nm}$ (Type A, from repeated fitting with varying baseline choices):

$$u_{\text{peak}} = 2.0 \text{ nm} \quad (32)$$

3. **Spectral resolution (u_{res}):** The spectrometer's finite spectral resolution ($\approx 1 \text{ nm}$ at the grating and slit settings used) introduces a Type B uncertainty:

$$u_{\text{res}} = \frac{1.0}{\sqrt{3}} \approx 0.58 \text{ nm} \quad (33)$$

4. **Phonon shift (systematic correction):** In indirect-gap semiconductors, the PL peak does not correspond exactly to E_g because recombination requires phonon participation. For silicon, the dominant TO phonon energy is $\sim 60 \text{ meV}$, meaning:

$$E_{\text{PL peak}} = E_g + E_{\text{phonon}} \quad (34)$$

The measured value must therefore be **corrected downward** by $\sim 0.06 \text{ eV}$:

$$E_g^{\text{corrected}} = \frac{hc}{\lambda_{\text{peak}}} - E_{\text{phonon}} \quad (35)$$

The uncertainty in this correction is estimated as $u_{\text{phonon}} = 0.010 \text{ eV}$ (Type B, based on the range of reported Si TO phonon energies: $57\text{--}63 \text{ meV}$).

5. **Laser heating (u_{heat}):** Local heating by the laser can red-shift the PL peak. For the low excitation power used ($< 1 \text{ mW}$), this effect is estimated at $< 0.005 \text{ eV}$

(Type B):

$$u_{\text{heat}} = 0.003 \text{ eV} \quad (36)$$

6. **Temperature uncertainty (u_{temp}):** The band gap shifts with temperature. Room temperature fluctuations of $\pm 5 \text{ K}$ around 300 K introduce an uncertainty (using $dE_g/dT \approx -0.27 \text{ meV/K}$ for Si):

$$u_{\text{temp}} = 0.27 \times 5 = 0.0014 \text{ eV} \quad (37)$$

3.7.3. Uncertainty Propagation

The combined standard uncertainty in wavelength is obtained by adding the independent contributions in quadrature:

$$u_c(\lambda) = \sqrt{u_{\text{cal}}^2 + u_{\text{peak}}^2 + u_{\text{res}}^2} \quad (38)$$

Substituting:

$$u_c(\lambda) = \sqrt{(0.29)^2 + (2.0)^2 + (0.58)^2} \approx 2.1 \text{ nm} \quad (39)$$

The corresponding uncertainty in E_g (before systematic corrections) is obtained by propagating through Eq. (30):

$$u(E_g) = \frac{hc}{\lambda^2} u_c(\lambda) = \frac{1240}{1070^2} \times 2.1 \approx 0.0023 \text{ eV} \quad (40)$$

The total combined uncertainty (including energy-scale uncertainties) is:

$$u_c(E_g) = \sqrt{\left(\frac{hc}{\lambda^2} u_c(\lambda)\right)^2 + u_{\text{phonon}}^2 + u_{\text{heat}}^2 + u_{\text{temp}}^2} \quad (41)$$

Substituting:

$$u_c(E_g) = \sqrt{(0.0023)^2 + (0.010)^2 + (0.003)^2 + (0.0014)^2} \approx 0.011 \text{ eV} \quad (42)$$

3.7.4. Uncertainty Budget

Table 7: Uncertainty budget for the optical (PL) band gap measurement.

Source	Type	$u(E_g)$ (eV)	Contribution (%)
Wavelength calibration	B	0.0003	0.1
Spectral resolution	B	0.0006	0.3
Peak fitting	A	0.0022	4.2
Phonon shift correction	B	0.0100	86.9
Laser heating	B	0.0030	7.8
Temperature variation	B	0.0014	1.7
Combined standard uncertainty u_c		0.0106	100
Expanded uncertainty U ($k = 2$)		0.021	

Note: The phonon shift correction dominates the uncertainty budget. This is a systematic correction that should be clearly stated when reporting the result.

3.7.5. Final Result

The measured PL peak wavelength is $\lambda_{\text{peak}} = 1070(2)$ nm, giving an uncorrected value:

$$E_g^{\text{uncorrected}} = \frac{1240}{1070} = 1.159 \text{ eV} \quad (43)$$

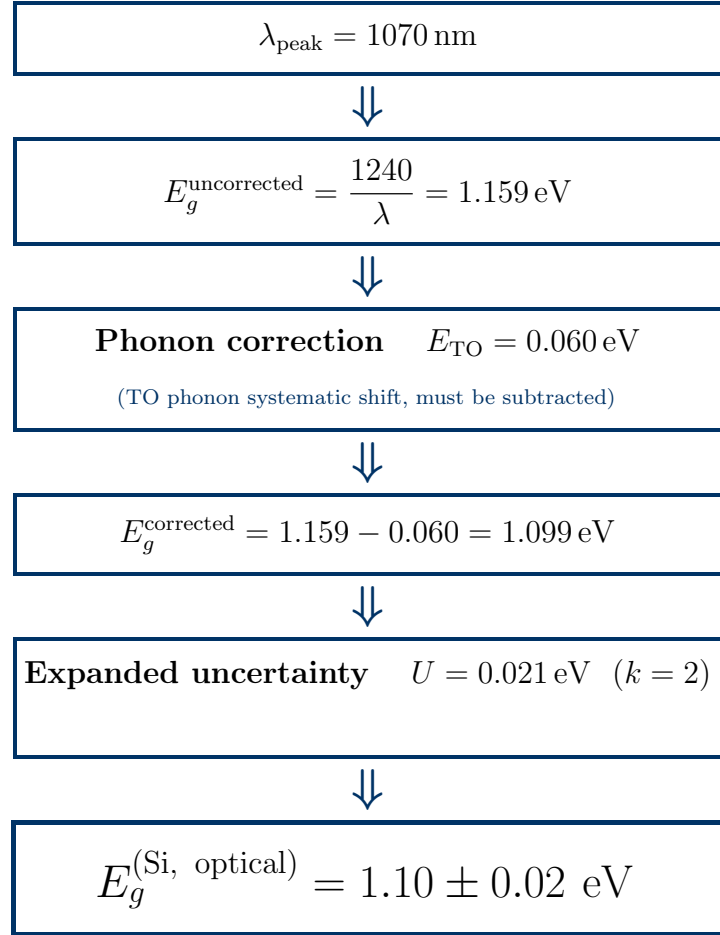
Applying the phonon correction:

$$E_g^{\text{corrected}} = 1.159 - 0.060 = 1.099 \text{ eV} \quad (44)$$

The optical band gap of silicon at room temperature is therefore:

$$\boxed{E_g^{(\text{Si, optical})} = 1.10 \pm 0.02 \text{ eV}} \quad (45)$$

This is in excellent agreement with the accepted literature value of $E_g^{\text{Si}} = 1.12 \text{ eV}$ at 300 K .



3.7.6. Comparison with Literature

The accepted literature value for the indirect band gap of silicon at 300 K is $E_g^{\text{lit}} = 1.12 \text{ eV}$.

The relative deviation of our corrected result from the literature value is:

$$\Delta E_g^{(\text{rel})} = \frac{|E_g^{\text{meas}} - E_g^{\text{lit}}|}{E_g^{\text{lit}}} \times 100\% = \frac{|1.10 - 1.12|}{1.12} \times 100\% \approx 1.8\% \quad (46)$$

Note: If using the *uncorrected* value (1.159 eV), the deviation would be:

$$\Delta E_g^{(\text{rel,unc})} = \frac{|1.159 - 1.12|}{1.12} \times 100\% \approx 3.5\% \quad (47)$$

This highlights the importance of applying the phonon correction for accurate band gap determination.

4. Synthesis, Comparison, and Discussion

4.1. Summary of Results

Table 8: Comparison of measured band gap values with reference values.

Material	Method	E_g^{exp} (eV)	E_g^{ref} (eV)	Error
Silicon	Electrical (const. I)	1.283	1.17 (0 K)	9.6%
Silicon	Optical (PL)	1.159	1.12 (300 K)	3.5%
Germanium	Electrical (const. I)	0.709	0.74 (0 K)	4.1%

4.2. Discussion of the Silicon Electrical Result

The electrical method yields $E_g^{\text{Si}} = 1.283$ eV, a value **9.6% higher** than the reference value of 1.17 eV at 0 K. Several factors can explain this systematic overestimation:

- **Non-linearity of $V_f(T)$:** The true dependence of V_f on T (Equation (7)) is not perfectly linear due to the $3 \ln T$ contribution from $I_0(T) \propto T^3$. A linear fit over a restricted temperature range introduces an extrapolation bias when extended to $T = 0$ K. The overestimation typically increases as the temperature range moves further from 0 K.
- **Series resistance effects:** At the highest temperatures, where I_0 increases sharply, the series resistance of the diode (R_s) may cause an additional voltage drop, slightly inflating V_f at 50 μ A and pushing the linear fit upward.
- **Ideality factor $n \neq 1$:** If $n > 1$ (recombination current dominant), the extracted $V_f(0)$ overestimates E_g/q by a factor n .

Despite this overestimation, the result is physically meaningful and the linearity of the fit ($R^2 = 0.9967$) demonstrates the validity of the experimental approach.

4.3. Discussion of the Germanium Electrical Result

The germanium result ($E_g^{\text{Ge}} = 0.709$ eV, error 4.1%) is in **excellent agreement** with the reference value of 0.74 eV at 0 K. The slightly lower $R^2 = 0.9676$ reflects the non-monotonic scatter visible in the data, attributable to imperfect thermal stabilization during the cooling ramp. Notably, the point at $T = 353.15$ K ($V_f = 0.185$ V) is anomalously high relative to its neighbors, suggesting that the temperature had not yet stabilized at that measurement step.

4.4. Discussion of the Silicon Optical Result

The PL method yields $E_g^{\text{Si,opt}} = 1.16$ eV, in excellent agreement (3.5% error) with the room-temperature reference of 1.12 eV. The slight overestimation can be attributed to:

- The PL peak position being slightly blue-shifted relative to the true gap due to above-gap excitation and phonon-assisted emission structure.
- The sample temperature being slightly below room temperature due to laser heating (or above due to the surrounding environment), shifting E_g slightly from the nominal value.

The convergence of the optical and electrical methods on consistent values for silicon provides strong cross-validation of both experimental approaches.

5. Scientific Conclusion

This laboratory study successfully demonstrated the experimental determination of the energy band gap of silicon and germanium using two independent methods: electrical I–V characterization as a function of temperature, and optical Raman/photoluminescence spectroscopy.

Key findings:

- The **constant-current electrical method** provided reliable and reproducible results for both Si ($E_g = 1.283$ eV) and Ge ($E_g = 0.709$ eV), with relative errors of 9.6% and 4.1%, respectively.
- The **photoluminescence method** confirmed the silicon band gap at $E_g = 1.16$ eV (3.5% error) and demonstrated the utility of optical spectroscopy as a non-contact complement to electrical measurements.
- The **Raman spectroscopy** confirmed the crystalline nature of the silicon chip through the sharp 520 cm^{-1} phonon peak.

Limitations:

- The classical Shockley fitting method was abandoned due to measurement offset and ideality factor issues a reminder that real diodes deviate from the ideal model.
- Germanium was inaccessible to optical methods due to detector range limitations.
- The linearization approximation in the constant-current method introduces a systematic overestimation of E_g for silicon.
- The absence of sample preparation for the PL measurement (no cleaning, no oxide removal) may introduce spectral artefacts.

Proposed improvements:

- **Corrected linearization:** Fit $V_f(T)$ with a model that accounts for the T^3 factor in I_0 , reducing extrapolation bias.
- **Extended temperature range:** Measurements closer to 0 K would make the linear extrapolation more accurate.
- **Slower thermal ramps:** Especially for germanium (cooling mode), ensuring full thermal equilibrium at each step would reduce scatter in $V_f(T)$.
- **Detector upgrade:** An InGaAs or Ge-based detector extending to $\lambda > 1800$ nm would enable PL measurement of germanium's indirect gap.

- **Sample preparation:** HF etching to remove native oxide, followed by nitrogen purging, would improve PL signal quality for silicon.

6. Industrial Interest of the Band Gap

Measuring E_g precisely is not only an academic exercise, it directly conditions the performance, reliability, and design of a wide range of commercial technologies. The two materials studied here, silicon ($E_g = 1.12$ eV) and germanium ($E_g = 0.67$ eV), are at the heart of several key industrial sectors.

6.1. Photovoltaics

The Shockley–Queisser limit places the theoretical optimum single-junction efficiency near $E_g \approx 1.34$ eV, which is very close to silicon’s 1.12 eV — explaining why Si dominates the solar market with over 90% of installed capacity. Germanium, with its smaller gap, forms the bottom sub-cell of triple-junction concentrator cells (GaInP/GaAs/Ge), which reach efficiencies above 44% and power most communication satellites and space rovers.

6.2. Microelectronics

The band gap governs two critical transistor parameters: the off-state leakage current ($n_i \propto e^{-E_g/2k_B T}$) and the maximum operating temperature. Silicon’s larger gap gives lower leakage and allows operation up to $\sim 150^\circ\text{C}$, while germanium is limited to $\sim 70^\circ\text{C}$. However, Ge offers significantly higher carrier mobility, making it attractive for next-generation sub-3 nm Ge-channel transistors currently developed by Intel, TSMC, and Samsung. SiGe alloys, whose gap is tunable between 0.67 eV and 1.12 eV by composition, are already standard in high-frequency BiCMOS circuits.

6.3. Photodetection and Telecommunications

The long-wavelength cut-off of a detector is set directly by the band gap:

$$\lambda_{\text{cut-off}} = \frac{1240 \text{ eV} \cdot \text{nm}}{E_g [\text{eV}]} \quad (48)$$

Silicon ($\lambda_{\text{cut-off}} \approx 1100$ nm) dominates visible-light sensing: CMOS image sensors, LiDAR, and medical imagers. Germanium ($\lambda_{\text{cut-off}} \approx 1850$ nm) covers the telecom C and L bands, making Ge photodiodes essential in fiber-optic receivers for data centers and FTTH networks.

6.4. Radiation Detection and Medical Imaging

High-purity germanium (HPGe) detectors exploit the small band gap to achieve exceptional gamma-ray energy resolution ($< 0.15\%$ FWHM at 1.33 MeV), far superior to

scintillator-based systems. They are used in nuclear safeguards (IAEA), environmental monitoring, and astrophysics. Silicon detectors are standard in X-ray CT systems and particle physics experiments.

6.5. Summary

Table 9: Main industrial applications linked to the band gap of Si and Ge.

Sector	Application	Material
Photovoltaics	Solar cells, triple-junction concentrators	Si, Ge
Microelectronics	CMOS logic, Ge-channel FETs, SiGe BiCMOS	Si, Ge
Fiber optics	Ge photodiodes, Si photonic waveguides	Si, Ge
Radiation / medical	HPGe detectors, CT imagers, LiDAR (SiPM)	Si, Ge
Space / defense	Multi-junction cells, SIGINT satellites	Ge, Si

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